

LOVELY PROFESSIONAL UNIVERSITY

Sample Question Set - Unit 2

Course Code: CSE252

Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Course Outcome: CO2

Paper: Sample Subjective Questions

Maximum Marks: 10 x 10 = 100

Unit: 2

Unit Title: Data Preparation and Machine Learning Workflow

Unit Mapped RBT: L3 (Apply)

Total Questions: 10

Suggested Duration: Not timed - practice set

Subjective Questions (10 x 10 = 100 Marks)

Q1. Apply an exploratory data analysis to a dataset you have just received. (10 Marks)

- (a) State the checks you would carry out and what each one would tell you.
- (b) Use two of those findings to explain what preparation the data would then need.

[L3 - Apply]

Q2. Compare mean, median and mode as values used to fill a missing entry. (10 Marks)

- (a) State the kind of column each of the three suits.
- (b) Use a column of annual incomes holding a few very large values to select one of the three, and justify it.

[L3 - Apply]

Q3. Examine the following workflow and correct it. An analyst encodes an unordered categorical column as 0, 1 and 2 and feeds it to a linear regression model. (10 Marks)

- (a) State what the model will read into those numbers that the data does not support.
- (b) Use the correct encoding to describe how the column should have been prepared, and what it costs.

[L3 - Apply]

Q4. Use cross validation to compare two candidate models on a dataset of moderate size. (10 Marks)

- (a) State how k-fold cross validation proceeds and what is reported at the end.
- (b) Use the results to explain how you would choose between the two, and what the test set is then for.

[L3 - Apply]

Q5. Apply feature engineering to a table of online orders holding a timestamp, an amount and a customer identifier. (10 Marks)

- (a) State three features you would construct and what each one would let the model see.
- (b) Use one of the three to explain why a well-chosen feature can matter more than the choice of algorithm.

[L3 - Apply]

Q6. Compare one-hot encoding with ordinal encoding of a categorical column. (10 Marks)

- (a) State what each of the two produces from the column.
- (b) Use a column of ordered sizes and a column of unordered cities to select an encoding for each, with reasons.

[L3 - Apply]

Q7. Examine a dataset in which two numeric features are found to be almost perfectly correlated. (10 Marks)

- (a) State what that correlation means for the information the two carry.
- (b) Use feature selection to describe what you would do, and state one situation in which you would keep both.

[L3 - Apply]

Q8. Apply the train, validation and test arrangement to a project of your own choosing. (10 Marks)

- (a) State what each of the three partitions is used for, and in what order.
- (b) Use this to explain what goes wrong when the test set is consulted repeatedly during development.

[L3 - Apply]

Q9. Use NumPy, Pandas and Matplotlib to describe how a new dataset is loaded and examined. (10 Marks)

- (a) State what each of the three libraries contributes at this stage.

(b) Use a named plot from Matplotlib to describe one thing you would check and what a bad result would look like.

[L3 - Apply]

Q10. Relate the choice of scaling method to the model that will be trained afterwards. (10 Marks)

(a) State which kinds of model require their features to be scaled and which do not, with the reason.

(b) Use a column holding a few extreme values to choose between standardisation, min-max scaling and a robust scaler.

[L3 - Apply]

----- End of Sample Question Set -----